Simulation Report

Experiment 1 - Hardware True Random Number Generator on the PYNQ-Z2

1. Setup

Item	Value
Simulator	Icarus Verilog 12 (iverilog -g2012, SystemVerilog)
Design under test	trng_top (RTL/*.sv)
Testbench	Testbench/trng_tb.sv
Waveform dump	Simulation/wave_dump.sv (records the main signals to trng_wave.vcd)
Clock	100 MHz (10 ns period), same as the board
Simulated time	100.5 us

2. What the testbench checks

The testbench drives the AXI4-Lite port of **trng_top** the same way the ARM processor does on the board. It uses two small tasks, *axi_write* and *axi_read*, which perform one complete AXI handshake each.

#	Step	Expected result
1	Read WINDOW register (0x0C) after reset	Reads 6 (the default window)
2	Write 0 to WINDOW and read it back	Reads 0, so the register can be written
3	Read DATA (0x08) before enabling	Returns 0 because the FIFO is empty
4	Write 1 to CTRL (0x00) and wait 100 us	Generator runs and fills the FIFO
5	Read STATUS (0x04)	FIFO not empty; LED1 lit
6	Read DATA twice	Two different 32-bit random words

Any failed check stops the simulation with *$fatal*, so reaching the PASS line means every check passed. The window is set to k = 0 (one raw bit per clock) only to keep the simulation short; the board uses k = 6.

Note on randomness in simulation: a simulator has no thermal noise, so the ring oscillators are modelled with fixed 1 ns gate delays and their outputs are periodic. The simulation therefore checks that the logic works (sampling, Von Neumann filter, FIFO, AXI registers, LEDs). The quality of the randomness can only be measured on the real FPGA, which is done in the hardware section of the main report.

3. Result

PASS - the TRNG produced random words over AXI. 78 words were waiting in the FIFO after 100 us and the two words read were 0x3A83365A and 0xCB930ACA.

The two "sorry: constant selects" lines are a known Icarus Verilog limitation message about the always_comb block in trng_axi_wrapper.sv; it does not change the result, and Vivado synthesises the same code without warnings.

Transcript

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Experiment 1 - Hardware True Random Number Generator (PYNQ-Z2)
Simulation transcript, 2026-09-18 14:55
Simulator: Icarus Verilog version 12.0 (devel) (s20150603-1539-g2693dd32b)

> iverilog -g2012 -o trng_sim.vvp -s trng_tb -s wave_dump ../RTL/*.sv ../Testbench/trng_tb.sv wave_dump.sv
(run inside Simulation/)
../RTL/trng_axi_wrapper.sv:114: sorry: constant selects in always_* processes are not currently supported (a
ll bits will be included).
../RTL/trng_axi_wrapper.sv:114: sorry: constant selects in always_* processes are not currently supported (a
ll bits will be included).

> vvp trng_sim.vvp
VCD info: dumpfile trng_wave.vcd opened for output.
status = 004e0000 (78 words waiting)
random words: 3a83365a cb930aca
PASS: TRNG produced random words over AXI
../Testbench/trng_tb.sv:97: $finish called at 100525000 (1ps)
```

4. Waveform

(a) Reset and AXI configuration: the window register is read (6), written with 0, then CTRL is written with 1 and *enable* goes high. (b) While running, the four ring-oscillator outputs are sampled, XORed into *raw_bit*, and the Von Neumann filter raises *valid* only when a pair of bits differs. After 32 valid bits *word_ready* pulses and the word 0x3A83365A enters the FIFO. (c) At the end the STATUS read returns 0x004E0000 (78 words) and the two DATA reads return the random words while the FIFO count drops. (d) The FIFO fills steadily during the run.



Figure 1 - Simulation waveform of trng_tb (drawn from trng_wave.vcd)